



LimeSDR-XTRX_v1.4

CONTROLLED IMPEDANCE

GENERAL PARAMETERS		
Parameter	Layer/Layers	Value
Copper foil thickness	Top	17.5um
Dielectric thickness between layers	Top-L2	102um (4 mils)
Dielectric permittivity between layers (Er)		4.2
Copper foil thickness	Bottom	17.5um
Dielectric thickness between layers	Bottom-L7	102um (4 mils)
Dielectric permittivity between layers (Er)		4.2

CALCULATIONS

RF (Top)	Target impedance	Single-ended	50R
	Additional comments	Top layer, no side GND plane	
	Top layer copper foil thickness	17.5um	
	Track width	0.18mm	
	Distance to reference plane	>3h	
	Dielectric thickness between layers	Top-L2	102um (4 mils)
	Dielectric permittivity between layers (Er)		4.2
	Approximate microstrip line impedance	50 Ohms (+/- 10% tolerance)	
RF (Top)	Target impedance	Differential	100R
	Additional comments	Top layer, no side GND plane	
	Top layer copper foil thickness	17.5um	
	Track width	0.1 mm (3.93 mils)	
	Track spacing	0.1 mm (3.93 mils)	
	Distance to reference plane	>3h	
	Dielectric thickness between layers	Top-L2	102um (4 mils)
	Dielectric permittivity between layers (Er)		4.2
	Approximate microstrip line impedance	100 Ohms (+/- 10% tolerance)	
RF (Bottom)	Target impedance	Single-ended	50R
	Additional comments	Bottom layer, no side GND plane	
	Top layer copper foil thickness	17.5 um	
	Track width	0.18mm	
	Distance to reference plane	>3h	
	Dielectric thickness between layers	Bottom-L7	102um (4 mils)
	Dielectric permittivity between layers (Er)		4.2
	Approximate microstrip line impedance	49.99 Ohms (+/- 10% tolerance)	
PCIe (Top and Bottom)	Target impedance	Differential	85R
	Additional comments	Top/Bottom layers, no side GND	
	Top layer copper foil thickness	17.5 um	
	Track width	0.14 mm (5.5 mils)	
	Track spacing	0.1 mm (3.93 mils)	
	Distance to reference plane	>3h	
	Dielectric thickness between layers	Top-L2 Bottom-L7	173um (6.8 mils)
	Dielectric permittivity between layers (Er)		4.2
	Approximate microstrip line impedance	85 Ohms (+/- 10% tolerance)	



STACKUP AND LAYER DESCRIPTION

GERBER LAYER NAMES:

GTP Top solder paste
GTO Silkscreen
GTS Soldermask (halogen free)
GTL 0.5oz+plating

G1 1oz (35um)
G2 1oz (35um)
G3 1oz (35um)
G4 1oz (35um)
G5 1oz (35um)
G6 1oz (35um)
GBL 0.5oz+plating
GBS Soldermask (halogen free)
GBO Silkscreen
GBP Bottom solder paste

Total PCB thickness: 1mm +/- 10%

Via type #2
Micro via
L7-Bot

Via type #1
Blind via
L5-Bot

TH via
Top-Bot

ELECTRICAL LAYERS:

Top: RF/Signal/PWR/GND

L2: GND

L3: Signal/PWR/GND

L4: Signal/PWR/GND

L5: GND

L6: CLK/PWR/GND

L7: GND

Bottom: RF/Signal/PWR/GND

ADDITIONAL LAYERS:

Mechanical 1: Board cutout
ASM TOP: Assembly top
ASM BOT: Assembly bottom
Mechanical 13: Component 3D body
Notes: Board shape and frame

VERY IMPORTANT NOTES

GENERAL PCB SPECS			ADDITIONAL REQUIREMENTS														
Minimum copper to copper spacing		0.1mm (3.9mil)		● All 0.2mm vias including 0.35mm ring and 0.2mm drill via-in-pads (IC1 and IC7) must be resin filled with metal cap.													
Minimum track width		0.1mm (3.9mil)															
PCB thickness		1mm +/-10%		IC1 thermal pad vias with 0.4mm ring and 0.2mm drill must be resin filled with metal cap. IC1 thermal pad vias with 0.5mm ring and 0.2mm drill must be left open (NO resin fill with metal cap). 4 vias in total, marked with note.													
PCB material		IT-180A preferred or equivalent RF-rated material (up to 3 GHz or higher)															
		External layer		0.5 oz+plating													
Copper weight		Internal layer		1 oz													
		DARK BLUE		● Electrical test : 100 % netlist.													
Solder mask		Both sides				● Boards are to be individually bagged.											
		Halogen free															
		Glossy finish (NOT matte)		PCB vendor to silkscreen UL and RoHS compliance marks, vendor logo and date code on bottom where shown (ignore if none of the info will be placed on PCB)													
Silkscreen		White epoxy ink															
		Both sides		Assembly note: Assembly house MUST provide notes in paper with shipped board if there were any changes during assembly and the board is not assembled 100% according to BOM and P&P files. Note example: ● <table><tr><th>Part</th><th>Initial BOM asm. note</th><th>Current PCB status</th><th>Comment</th></tr><tr><td>R1</td><td>FIT</td><td>NF</td><td>Not mounted due to bad footprint</td></tr><tr><td>IC5</td><td>FIT</td><td>NF</td><td>Not mounted due to part shortage</td></tr></table>		Part	Initial BOM asm. note	Current PCB status	Comment	R1	FIT	NF	Not mounted due to bad footprint	IC5	FIT	NF	Not mounted due to part shortage
		Part	Initial BOM asm. note			Current PCB status	Comment										
R1	FIT	NF	Not mounted due to bad footprint														
IC5	FIT	NF	Not mounted due to part shortage														
Halogen free																	
No silkscreen on pads																	
Hole types on the PCB and information		Plated ☒		Non-plated ☒													
		Hole diameters are final ● manufactured diameters INCLUDING HOLE METALIZATION.															
Route process		U-score ☐		Tab route ☒													
		U-score and tab route ☐		● PCIe REQUIREMENT #1: PCI EXPRESS pads (top and bottom) - 30 micro inches of gold over 50 micro inches nickel plating													
Panel		Yes ☐				No ☒											
Surface finish (both sides)		HASL lead free		☐													
		HASL with lead		☐													
		Immersion gold		0.05-0.10um of gold over 2.50-5.00um of nickel ☒													
		OSP		☐													
		Hard gold		PCIe fingers ☒													
				● Blind vias can be coated with copper													

PCI connector
edge chamfer ← PCIe REQUIREMENTS #2 AND #3

45 (+/- 7)
degrees

0.25mm

1mm +/- 0.1mm

PCIe REQUIREMENT #1

0,25cm

2,42cm

2x R=1,30mm

2,97cm
5,61cm

5,08cm

6,40cm

